

topologies have the same compact sets, they have the same singular simplices and hence the same singular homology groups.

Let $C = C_*(X; R)$ and $C' = C_*(Y; R)$, the cellular chain complexes with coefficients in R . Then $C \otimes_R C' = C_*(X \times Y; R)$ by Proposition 3B.1, so the algebraic Künneth formula gives the desired short exact sequences. Their naturality follows from naturality in the algebraic Künneth formula, since we can homotope arbitrary maps $X \rightarrow X'$ and $Y \rightarrow Y'$ to be cellular by Theorem 4.8, assuring that they induce chain maps of cellular chain complexes. $\square$

With field coefficients the Künneth formula simplifies because the Tor terms are always zero over a field:

Corollary 3B.7. *If F is a field and X and Y are CW complexes, then the cross product map $h: \bigoplus_i (H_i(X; F) \otimes_F H_{n-i}(Y; F)) \rightarrow H_n(X \times Y; F)$ is an isomorphism for all n . $\square$*

There is also a relative version of the Künneth formula for CW pairs (X, A) and (Y, B) . This is a split short exact sequence

$$0 \rightarrow \bigoplus_i (H_i(X, A; R) \otimes_R H_{n-i}(Y, B; R)) \rightarrow H_n(X \times Y, A \times Y \cup X \times B; R) \rightarrow \bigoplus_i \text{Tor}_R(H_i(X, A; R), H_{n-i-1}(Y, B; R)) \rightarrow 0$$

for R a principal ideal domain. This too follows from the algebraic Künneth formula since the isomorphism of cellular chain complexes $C_*(X \times Y) \approx C_*(X) \otimes C_*(Y)$ passes down to a quotient isomorphism

$$C_*(X \times Y) / C_*(A \times Y \cup X \times B) \approx C_*(X) / C_*(A) \otimes C_*(Y) / C_*(B)$$

since bases for these three relative cellular chain complexes correspond bijectively with the cells of $(X - A) \times (Y - B)$, $X - A$, and $Y - B$, respectively.

As a special case, suppose A and B are basepoints $x_0 \in X$ and $y_0 \in Y$. Then the subcomplex $A \times Y \cup X \times B$ can be identified with the wedge sum $X \vee Y$ and the quotient $X \times Y / X \vee Y$ is the smash product $X \wedge Y$. Thus we have a reduced Künneth formula

$$0 \rightarrow \bigoplus_i (\tilde{H}_i(X; R) \otimes_R \tilde{H}_{n-i}(Y; R)) \rightarrow \tilde{H}_n(X \wedge Y; R) \rightarrow \bigoplus_i \text{Tor}_R(\tilde{H}_i(X; R), \tilde{H}_{n-i-1}(Y; R)) \rightarrow 0$$

If we take $Y = S^k$ for example, then $X \wedge S^k$ is the k -fold reduced suspension of X , and we obtain isomorphisms $\tilde{H}_n(X; R) \approx \tilde{H}_{n+k}(X \wedge S^k; R)$.

The Künneth formula and the universal coefficient theorem can be combined to give a more concise formula $H_n(X \times Y; R) \approx \bigoplus_i H_i(X; H_{n-i}(Y; R))$. The naturality of this isomorphism is somewhat problematic, however, since it uses the splittings in the Künneth formula and universal coefficient theorem. With a little more algebra the formula can be shown to hold more generally for an arbitrary coefficient group G in place of R ; see [Hilton & Wylie 1967], p. 227.

There is an analogous formula $\tilde{H}_n(X \wedge Y; R) \approx \bigoplus_i \tilde{H}_i(X; \tilde{H}_{n-i}(Y; R))$. As a special case, when Y is a Moore space $M(G, k)$ we obtain isomorphisms $\tilde{H}_n(X; G) \approx \tilde{H}_{n+k}(X \wedge M(G, k); \mathbb{Z})$. Again naturality is an issue, but in this case there is a natural isomorphism obtainable by applying Theorem 4.59 in §4.3, after verifying that the functors $h_n(X) = \tilde{H}_{n+k}(X \wedge M(G, k); \mathbb{Z})$ define a reduced homology theory, which is not hard. The isomorphism $\tilde{H}_n(X; G) \approx \tilde{H}_{n+k}(X \wedge M(G, k); \mathbb{Z})$ says that homology with arbitrary coefficients can be obtained from homology with $\mathbb{Z}$ coefficients by a topological construction as well as by the algebra of tensor products. For general homology theories this formula can be used as a definition of homology with coefficients.

One might wonder about a cohomology version of the Künneth formula. Taking coefficients in a field F and using the natural isomorphism $\text{Hom}(A \otimes B, C) \approx \text{Hom}(A, \text{Hom}(B, C))$, the Künneth formula for homology and the universal coefficient theorem give isomorphisms

$$\begin{aligned} H^n(X \times Y; F) &\approx \text{Hom}_F(H_n(X \times Y; F), F) \approx \bigoplus_i \text{Hom}_F(H_i(X; F) \otimes H_{n-i}(Y; F), F) \\ &\approx \bigoplus_i \text{Hom}_F(H_i(X; F), \text{Hom}_F(H_{n-i}(Y; F), F)) \\ &\approx \bigoplus_i \text{Hom}_F(H_i(X; F), H^{n-i}(Y; F)) \\ &\approx \bigoplus_i H^i(X; H^{n-i}(Y; F)) \end{aligned}$$

More generally, there are isomorphisms $H^n(X \times Y; G) \approx \bigoplus_i H^i(X; H^{n-i}(Y; G))$ for any coefficient group G ; see [Hilton & Wylie 1967], p. 227. However, in practice it usually suffices to apply the Künneth formula for homology and the universal coefficient theorem for cohomology separately. Also, Theorem 3.16 shows that with stronger hypotheses one can draw stronger conclusions using cup products.

The Simplicial Cross Product

Let us sketch how the cross product $H_m(X; R) \otimes H_n(Y; R) \rightarrow H_{m+n}(X \times Y; R)$ can be defined directly in terms of singular homology. What one wants is a cross product at the level of singular chains, $C_m(X; R) \otimes C_n(Y; R) \rightarrow C_{m+n}(X \times Y; R)$. If we are given singular simplices $f: \Delta^m \rightarrow X$ and $g: \Delta^n \rightarrow Y$, then we have the product map $f \times g: \Delta^m \times \Delta^n \rightarrow X \times Y$, and the idea is to subdivide $\Delta^m \times \Delta^n$ into simplices of dimension $m + n$ and then take the sum of the restrictions of $f \times g$ to these simplices, with appropriate signs.

In the special cases that m or n is 1 we have already seen how to subdivide $\Delta^m \times \Delta^n$ into simplices when we constructed prism operators in §2.1. The generalization to $\Delta^m \times \Delta^n$ is not completely obvious, however. Label the vertices of Δ^m as $v_0, v_1, \dots, v_m$ and the vertices of Δ^n as $w_0, w_1, \dots, w_n$. Think of the pairs (i, j) with $0 \leq i \leq m$ and $0 \leq j \leq n$ as the vertices of an $m \times n$ rectangular grid in $\mathbb{R}^2$. Let σ be a path formed by a sequence of $m + n$ horizontal and vertical edges in this grid starting at $(0, 0)$ and ending at (m, n) , always moving either to the right or upward. To such a path σ we associate a linear map $\ell_\sigma: \Delta^{m+n} \rightarrow \Delta^m \times \Delta^n$ sending the k^{th} vertex of Δ^{m+n} to (v_{i_k}, w_{j_k}) where (i_k, j_k) is the k^{th} vertex of the edgepath σ . Then

we define a simplicial cross product

$$C_m(X; R) \otimes C_n(Y; R) \xrightarrow{\times} C_{m+n}(X \times Y; R)$$

by the formula

$$f \times g = \sum_{\sigma} (-1)^{|\sigma|} (f \times g) \ell_{\sigma}$$

where $|\sigma|$ is the number of squares in the grid lying below the path σ . Note that the symbol ' $\times$ ' means different things on the two sides of the equation. From this definition it is a calculation to show that $\partial(f \times g) = \partial f \times g + (-1)^m f \times \partial g$. This implies that the cross product of two cycles is a cycle, and the cross product of a cycle and a boundary is a boundary, so there is an induced cross product in singular homology.

One can see that the images of the maps ℓ_{σ} give a simplicial structure on $\Delta^m \times \Delta^n$ in the following way. We can view Δ^m as the subspace of $\mathbb{R}^m$ defined by the inequalities $0 \leq x_1 \leq \cdots \leq x_m \leq 1$, with the vertex v_i as the point having coordinates $m - i$ zeros followed by i ones. Similarly we have $\Delta^n \subset \mathbb{R}^n$ with coordinates $0 \leq y_1 \leq \cdots \leq y_n \leq 1$. The product $\Delta^m \times \Delta^n$ then consists of $(m + n)$ -tuples $(x_1, \dots, x_m, y_1, \dots, y_n)$ satisfying both sets of inequalities. The combined inequalities $0 \leq x_1 \leq \cdots \leq x_m \leq y_1 \leq \cdots \leq y_n \leq 1$ define a simplex Δ^{m+n} in $\Delta^m \times \Delta^n$, and every other point of $\Delta^m \times \Delta^n$ satisfies a similar set of inequalities obtained from $0 \leq x_1 \leq \cdots \leq x_m \leq y_1 \leq \cdots \leq y_n \leq 1$ by a permutation of the variables 'shuffling' the y_j 's into the x_i 's. Each such shuffle corresponds to an edgepath σ consisting of a rightward edge for each x_i and an upward edge for each y_j in the shuffled sequence. Thus we have $\Delta^m \times \Delta^n$ expressed as the union of simplices Δ_{σ}^{m+n} indexed by the edgepaths σ . One can check that these simplices fit together nicely to form a Δ -complex structure on $\Delta^m \times \Delta^n$, which is also a simplicial complex structure. See [Eilenberg & Steenrod 1952], p. 68. In fact this construction is sufficiently natural to make the product of any two Δ -complexes into a Δ -complex.

The Cohomology Cross Product

In §3.2 we defined a cross product

$$H^k(X; R) \times H^{\ell}(Y; R) \xrightarrow{\times} H^{k+\ell}(X \times Y; R)$$

in terms of the cup product. Let us now describe the alternative approach in which the cross product is defined directly via cellular cohomology, and then cup product is defined in terms of this cross product.

The cellular definition of cohomology cross product is very much like the definition in homology. Given CW complexes X and Y , define a cross product of cellular cochains $\varphi \in C^k(X; R)$ and $\psi \in C^{\ell}(Y; R)$ by setting

$$(\varphi \times \psi)(e_{\alpha}^k \times e_{\beta}^{\ell}) = \varphi(e_{\alpha}^k) \psi(e_{\beta}^{\ell})$$

and letting $\varphi \times \psi$ take the value 0 on $(k + \ell)$ -cells of $X \times Y$ which are not the product of a k -cell of X with an ℓ -cell of Y . Another way of saying this is to use the convention

that a cellular cochain in $C^k(X; R)$ takes the value 0 on cells of dimension different from k , and then we can let $(\varphi \times \psi)(e_\alpha^m \times e_\beta^n) = \varphi(e_\alpha^m)\psi(e_\beta^n)$ for all m and n .

The cellular coboundary formula $\delta(\varphi \times \psi) = \delta\varphi \times \psi + (-1)^k \varphi \times \delta\psi$ for cellular cochains $\varphi \in C^k(X; R)$ and $\psi \in C^\ell(Y; R)$ follows easily from the corresponding boundary formula in Proposition 3B.1, namely

$$\begin{aligned} \delta(\varphi \times \psi)(e_\alpha^m \times e_\beta^n) &= (\varphi \times \psi)(\partial(e_\alpha^m \times e_\beta^n)) \\ &= (\varphi \times \psi)(\partial e_\alpha^m \times e_\beta^n + (-1)^m e_\alpha^m \times \partial e_\beta^n) \\ &= \delta\varphi(e_\alpha^m)\psi(e_\beta^n) + (-1)^m \varphi(e_\alpha^m)\delta\psi(e_\beta^n) \\ &= (\delta\varphi \times \psi + (-1)^k \varphi \times \delta\psi)(e_\alpha^m \times e_\beta^n) \end{aligned}$$

where the coefficient $(-1)^m$ in the next-to-last line can be replaced by $(-1)^k$ since $\varphi(e_\alpha^m) = 0$ unless $k = m$. From the formula $\delta(\varphi \times \psi) = \delta\varphi \times \psi + (-1)^k \varphi \times \delta\psi$ it follows just as for homology and for cup product that there is an induced cross product in cellular cohomology.

To show this agrees with the earlier definition, we can first reduce to the case that X has trivial $(k-1)$ -skeleton and Y has trivial $(\ell-1)$ -skeleton via the commutative diagram

$$\begin{array}{ccc} H^k(X/X^{k-1}; R) \times H^\ell(Y/Y^{\ell-1}; R) & \xrightarrow{\times} & H^{k+\ell}(X/X^{k-1} \times Y/Y^{\ell-1}; R) \\ \downarrow & & \downarrow \\ H^k(X; R) \times H^\ell(Y; R) & \xrightarrow{\times} & H^{k+\ell}(X \times Y; R) \end{array}$$

The left-hand vertical map is surjective, so by commutativity, if the two definitions of cross product agree in the upper row, they agree in the lower row. Next, assuming X^{k-1} and $Y^{\ell-1}$ are trivial, consider the commutative diagram

$$\begin{array}{ccc} H^k(X; R) \times H^\ell(Y; R) & \xrightarrow{\times} & H^{k+\ell}(X \times Y; R) \\ \downarrow & & \downarrow \\ H^k(X^k; R) \times H^\ell(Y^\ell; R) & \xrightarrow{\times} & H^{k+\ell}(X^k \times Y^\ell; R) \end{array}$$

The vertical maps here are injective, $X^k \times Y^\ell$ being the $(k+\ell)$ -skeleton of $X \times Y$, so it suffices to see that the two definitions agree in the lower row. We have $X^k = \bigvee_\alpha S_\alpha^k$ and $Y^\ell = \bigvee_\beta S_\beta^\ell$, so by restriction to these wedge summands the question is reduced finally to the case of a product $S_\alpha^k \times S_\beta^\ell$. In this case, taking $R = \mathbb{Z}$, we showed in Theorem 3.16 that the cross product in question is the map $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ sending $(1, 1)$ to ± 1 , with the original definition of cross product. The same is obviously true using the cellular cross product. So for $R = \mathbb{Z}$ the two cross products agree up to sign, and it follows that this is also true for arbitrary R . We leave it to the reader to sort out the matter of signs.

To relate cross product to cup product we use the diagonal map $\Delta: X \rightarrow X \times X$, $x \mapsto (x, x)$. If we are given a definition of cross product, we can define cup product as the composition

$$H^k(X; R) \times H^\ell(X; R) \xrightarrow{\times} H^{k+\ell}(X \times X; R) \xrightarrow{\Delta^*} H^{k+\ell}(X; R)$$

This agrees with the original definition of cup product since we have $\Delta^*(a \times b) = \Delta^*(p_1^*(a) \smile p_2^*(b)) = \Delta^*(p_1^*(a)) \smile \Delta^*(p_2^*(b)) = a \smile b$, as both compositions $p_1\Delta$ and $p_2\Delta$ are the identity map of X .

Unfortunately, the definition of cellular cross product cannot be combined with Δ to give a definition of cup product at the level of cellular cochains. This is because Δ is not a cellular map, so it does not induce a map of cellular cochains. It is possible to homotope Δ to a cellular map by Theorem 4.8, but this involves arbitrary choices. For example, the diagonal of a square can be pushed across either adjacent triangle. In particular cases one might hope to understand the geometry well enough to compute an explicit cellular approximation to the diagonal map, but usually other techniques for computing cup products are preferable.

The cohomology cross product satisfies the same commutativity relation as for homology, namely $T^*(a \times b) = (-1)^{k\ell} b \times a$ for $T: X \times Y \rightarrow Y \times X$ the transposition map, $a \in H^k(Y; R)$, and $b \in H^\ell(X; R)$. The proof is the same as for homology. Taking $X = Y$ and noting that $T\Delta = \Delta$, we obtain a new proof of the commutativity property of cup product.

Exercises

1. Compute the groups $H_i(\mathbb{RP}^m \times \mathbb{RP}^n; G)$ and $H^i(\mathbb{RP}^m \times \mathbb{RP}^n; G)$ for $G = \mathbb{Z}$ and $\mathbb{Z}_2$ via the cellular chain and cochain complexes. [See Example 3B.4.]
2. Let C and C' be chain complexes, and let I be the chain complex consisting of $\mathbb{Z}$ in dimension 1 and $\mathbb{Z} \times \mathbb{Z}$ in dimension 0, with the boundary map taking a generator e in dimension 1 to the difference $v_1 - v_0$ of generators v_i of the two $\mathbb{Z}$'s in dimension 0. Show that a chain map $f: I \otimes C \rightarrow C'$ is precisely the same as a chain homotopy between the two chain maps $f_i: C \rightarrow C'$, $c \mapsto f(v_i \otimes c)$, $i = 0, 1$. [The chain homotopy is $h(c) = f(e \otimes c)$.]
3. Show that the splitting in the topological Künneth formula cannot be natural by considering the map $f \times \mathbb{1}: M(\mathbb{Z}_m, n) \times M(\mathbb{Z}_m, n) \rightarrow S^{n+1} \times M(\mathbb{Z}_m, n)$ where f collapses the n -skeleton of $M(\mathbb{Z}_m, n) = S^n \cup e^{n+1}$ to a point.
4. Show that the cross product of fundamental classes for closed R -orientable manifolds M and N is a fundamental class for $M \times N$.
5. Show that **slant products**

$$H_n(X \times Y; R) \times H^j(Y; R) \rightarrow H_{n-j}(X; R), \quad (e^i \times e^j, \varphi) \mapsto \varphi(e^j)e^i$$

$$H^n(X \times Y; R) \times H_j(Y; R) \rightarrow H^{n-j}(X; R), \quad (\varphi, e^j) \mapsto (e^i \mapsto \varphi(e^i \times e^j))$$

can be defined via the indicated cellular formulas. [These 'products' are in some ways more like division than multiplication, and this is reflected in the common notation a/b for them, or $a \setminus b$ when the order of the factors is reversed. The first of the two slant products is related to cap product in the same way that the cohomology cross product is related to cup product.]